

Matched WSI–RNA PAM50 Subtype Classification on TCGA-BRCA

1 Scope

This report summarises only the TCGA-BRCA PAM50 experiments run on the matched WSI–RNA cohort. Every result below uses the same four-class slide-level cohort of **986** H&E whole-slide images with available UNI2-h WSI embeddings and matched RNA-seq features. Samples labelled Normal-like and slides without matched paired modality data are excluded from this evaluation.

The goal is to compare the existing histology branch, a matched RNA-seq branch, and WSI–RNA fusion strategies under one consistent evaluation protocol.

2 Cohort and Task

The prediction target is four-class PAM50 molecular subtype classification. The experimental unit is a slide. Slides are matched to RNA-seq samples by TCGA case identifier, and labels are checked for consistency before training or evaluation.

Table 1: Matched WSI–RNA cohort used for all reported experiments.

Class index	PAM50 subtype	Slides
0	Luminal A	508
1	Luminal B	222
2	Basal-like	173
3	HER2-enriched	83
Total		986

3 Methods

3.1 Shared Evaluation Protocol

All models are evaluated with the same patient-level 10-fold split protocol. A patient appears in exactly one held-out test fold, so concatenating the ten fold-level test predictions gives one held-out prediction for every slide in the 986-slide cohort. Hyperparameters are selected on validation data only, and test metrics are reported after the validation choices are fixed.

For RNA-containing models, gene selection and normalisation are fitted inside each training fold only. The selected RNA representation uses the 10,000 highest-variance genes from the training split, then applies training-fold mean and standard deviation to validation and test samples. This avoids leakage from validation or test RNA distributions.

The primary metrics are macro AUC, accuracy, balanced accuracy, macro F1, and weighted F1. Macro AUC is computed one-vs-rest and averaged across classes. Balanced accuracy and macro F1 are included because the class distribution is imbalanced.

3.2 Model Families

WSI-only baseline. The histology branch is CLAM-MB trained on UNI2-h slide features. This branch predicts PAM50 subtype from H&E morphology alone.

Matched RNA-only baseline. The RNA branch is a multilayer perceptron using the fold-local selected gene features. The selected architecture uses hidden dimensions [1024, 512], dropout 0.4, AdamW, learning rate 0.000197465, weight decay 0.000700548, label smoothing 0.05, weighted loss, weighted sampling, batch size 64, and early stopping.

WSI-RNA fusion models. Each fusion model pairs a CLAM-MB WSI branch with an RNA MLP branch:

- **Concat fusion:** concatenate the CLAM slide embedding and RNA embedding, then classify through a hidden fusion head.
- **Gated fusion:** project WSI and RNA embeddings to a shared hidden space, learn a sigmoid gate from both embeddings, and classify the gated mixture.
- **Late probability fusion:** linearly combine RNA and WSI class probabilities, with the RNA/WSI weight selected on validation data.
- **Residual fusion:** keep the RNA prediction as the base and train a conservative WSI-conditioned residual correction.
- **Selective ensemble:** temperature-calibrate RNA, WSI, concat, gated, and residual probabilities on validation data, then combine them with fold-specific convex weights selected by validation balanced accuracy.

4 Results

4.1 Aggregate Held-out Test Performance

Table 2 reports metrics after concatenating held-out test predictions across all ten folds. All rows use the same 986-slide cohort.

Table 2: Aggregate held-out test results on the matched 986-slide cohort.

Method	n	Macro AUC $\uparrow$	Acc. $\uparrow$	Bal. Acc. $\uparrow$	Macro F1 $\uparrow$	Weighted F1 $\uparrow$	ECE $\downarrow$
WSI only	986	0.8879	0.7181	–	0.6500	0.7179	–
Matched RNA only	986	0.9785	0.8783	0.8809	0.8644	0.8799	–
WSI + RNA concat	986	0.9765	0.8884	0.8745	0.8653	0.8896	–
WSI + RNA gated	986	0.9744	0.8874	0.8724	0.8667	0.8885	–
Late probability fusion	986	0.9789	0.8803	0.8756	0.8657	0.8812	–
WSI + RNA residual	986	0.9787	0.8803	0.8535	0.8575	0.8797	–
Selective ensemble	986	0.9796	0.8996	0.8856	0.8804	0.8999	0.0256

The matched RNA-only branch is the strongest single-modality model, which is expected because PAM50 is defined from gene-expression patterns. Feature-level fusion improves accuracy over the matched RNA baseline, with concat reaching 0.8884 and gated fusion reaching 0.8874. The best overall result is the calibrated selective ensemble, which reaches 0.8996 accuracy, 0.9796 macro AUC, 0.8856 balanced accuracy, 0.8804 macro F1, and 0.8999 weighted F1.

4.2 Fold-level Stability

Table 3 reports mean and standard deviation across the ten held-out test folds. These fold-level values are complementary to Table 2; small differences are expected because aggregate metrics are computed after concatenating all held-out predictions.

Table 3: Fold-level held-out test performance on the matched 986-slide cohort.

Method	Test macro AUC $\uparrow$	Test accuracy $\uparrow$
Matched RNA only	0.9806 $\pm$ 0.0075	0.8787 $\pm$ 0.0285
WSI + RNA concat	0.9783 $\pm$ 0.0062	0.8890 $\pm$ 0.0276
WSI + RNA gated	0.9751 $\pm$ 0.0065	0.8879 $\pm$ 0.0282
Late probability fusion	0.9806 $\pm$ 0.0067	0.8806 $\pm$ 0.0268
WSI + RNA residual	0.9799 $\pm$ 0.0090	0.8804 $\pm$ 0.0286

4.3 Selective Ensemble Per-class Results

The selective ensemble is the strongest aggregate model. Its per-class performance is shown in Table 4.

Table 4: Selective ensemble per-class held-out test metrics.

Class	AUC $\uparrow$	Precision $\uparrow$	Recall $\uparrow$	F1 $\uparrow$	Support
Luminal A	0.9765	0.9306	0.9232	0.9269	508
Luminal B	0.9604	0.8145	0.8108	0.8126	222
Basal-like	0.9986	0.9826	0.9769	0.9797	173
HER2-enriched	0.9829	0.7753	0.8313	0.8023	83

Basal-like is the most separable class, while HER2-enriched and Luminal B remain the most challenging. The selective ensemble mainly improves the minority-class balance without sacrificing the dominant Luminal A class.

4.4 Gated Fusion Behaviour

For gated fusion, the aggregate mean WSI gate is 0.4709 and the mean RNA gate is 0.5291. Mean WSI gate values are similar across subtypes: 0.4661 for Luminal A, 0.4793 for Luminal B, 0.4739 for Basal-like, and 0.4710 for HER2-enriched. Correct predictions have mean WSI gate 0.4703, while incorrect predictions have 0.4757. In this run, the learned gate does not act as a strong confidence switch; it behaves more like a stable soft mixture that slightly favours RNA.

5 Interpretation

On the matched 986-slide cohort, RNA-seq provides substantially stronger PAM50 signal than histology alone. Direct feature-level fusion recovers the RNA-only performance and modestly improves accuracy, but it does not consistently improve macro AUC. The selective ensemble gives the best current result because it uses validation-calibrated modality weighting rather than forcing a single fused representation for every slide.

These results support using RNA as the molecular reference branch and WSI as a complementary source of morphological signal. For documentation and comparison tables, the most defensible headline result from this set of experiments is the calibrated selective ensemble on 986 matched slides: accuracy 0.8996, macro AUC 0.9796, balanced accuracy 0.8856, macro F1 0.8804, weighted F1 0.8999, and ECE 0.0256.

6 Related Work Context

Hezil et al. [1,2] report strong TCGA-BRCA multimodal PAM50 results using RNA-seq, CTransPath WSI features, several fusion mechanisms, and confidence-aware routing. Their results motivate the concat, gated, and selective-fusion strategies evaluated here, but the numbers are not directly comparable because cohort construction, feature extraction, split protocol, and reporting details differ. Prior multiomics and histopathology studies also support the observation that transcriptomic modalities carry strong PAM50 signal and that histology can add complementary information [3,4,5].

References

1. Hezil, N. et al. Selective multimodal deep learning for reliable breast cancer subtype classification from histopathology and genomic data. *Medical Engineering & Physics* (2025). [PubMed record](#).
2. Hezil, N. et al. Selective multimodal breast cancer subtype classification implementation and reported metrics. [Code repository](#).

3. Ektefaie, Y. et al. Integrative multiomics-histopathology analysis for breast cancer classification. *npj Breast Cancer* 7, 147 (2021). [Article](#).
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